

02-womens-history-stem-challenges

Women's History Month STEM Challenge Cards — Full Content

Visual Design Notes

- **Color Palette:** Vibrant purple, teal, hot pink, electric yellow on white backgrounds
 - **Style:** Modern, bold, “science lab” aesthetic — beakers, gears, stars, circuit patterns
 - **Card Layout:** 5.5” x 8.5” (half-page) cards — front has bio + photo area, back has challenge
 - **Each card has a unique accent color** while maintaining the overall theme
 - **Fonts:** Bold rounded sans-serif headers, clean body text
 - **Icons:** Custom STEM icons (beaker, rocket, calculator, gear, leaf, computer) on each card
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STEM Challenge Card 1: Mae Jemison — Parachute Drop Challenge

Biography Side

Mae C. Jemison (born 1956) became the first Black woman to travel to space when she flew aboard the Space Shuttle Endeavour in 1992. Before becoming an astronaut, she was a medical doctor and Peace Corps volunteer. Mae studied chemical engineering at Stanford and medicine at Cornell. She has said, “Never limit yourself because of others’ limited imagination.”

Challenge Side



The Parachute Drop Challenge

Your Mission: Design a parachute that will carry a small action figure or toy safely to the ground with the SLOWEST fall time.

Materials: - Plastic bags, tissue paper, or fabric scraps (various sizes) - String or yarn (30 cm pieces) - Tape - Small toy figure or weight (same weight for all teams) - Stopwatch - Meter stick (to measure drop height)

Steps: 1. **Research (5 min):** Look at your materials. How does a parachute slow things down? (Air resistance!) 2. **Plan (5 min):** Sketch your parachute design. Label the parts. 3. **Build (15 min):** Construct your parachute. 4. **Test (10 min):** Drop from the same height 3 times. Record fall time each trial. 5. **Improve (10 min):** Make ONE change to improve your design. Test again.

Record: Trial times, what you changed, and why.

TEKS Connection: Sci 4.1(A) Ask questions about the natural world; Sci 4.1(B) Design simple experimental investigations; Math 4.8(C) Solve problems using data from tables/graphs.

STEM Challenge Card 2: Ellen Ochoa — Robotic Arm Challenge

Biography Side

Ellen Ochoa (born 1958) was the first Hispanic woman to go to space in 1993. She flew on four space shuttle missions and later became the director of NASA's Johnson Space Center in Houston, Texas! Ellen is also an inventor — she holds patents for optical systems used in space. She grew up in California and earned her doctorate in electrical engineering from Stanford.

Challenge Side

The Robotic Arm Challenge

Your Mission: Build a “robotic arm” from simple materials that can pick up and move 3 small objects from one cup to another.

Materials: - Cardboard strips, craft sticks, or straws - Binder clips, clothespins, or rubber bands - String or yarn - Tape and glue - 2 cups - Small objects to move (cotton balls, erasers, paper clips)

Steps: 1. **Explore (5 min):** How does a real robotic arm work? It has joints and a gripper! 2. **Plan (5 min):** Sketch your arm. How will it grip? How will it reach? 3. **Build (20 min):** Construct your robotic arm. It must be at least 12 inches long. 4. **Test (10 min):** Try to move 3 objects. Time yourself. Can you do it in under 60 seconds? 5. **Improve (10 min):** Redesign one part to work better.

Rules: You can only touch the arm at its base — no touching the gripper end directly!

TEKS Connection: Sci 3.1(A), 5.1(A) Plan and implement investigations; Math 3.7(A) Represent fractions with objects.

STEM Challenge Card 3: Katherine Johnson — Measurement & Calculation Challenge

Biography Side

Katherine Johnson (1918-2020) was a brilliant mathematician who calculated the flight paths for NASA's first human spaceflights. Her math was so accurate that astronaut John Glenn refused to fly until Katherine personally verified the computer's calculations. She was featured in the movie *Hidden Figures* and received the Presidential Medal of Freedom.

Challenge Side

The Trajectory Calculation Challenge

Your Mission: Use measurement and math to predict where a marble will land when rolled off a ramp at different heights.

Materials: - Ruler and measuring tape - Marble - Books or blocks (to build ramps at different heights) - A flat board or stiff cardboard (ramp surface) - Masking tape (to mark landing spots) - Recording sheet

Steps: 1. **Set up:** Build a ramp using 2 books. Roll the marble off the edge of a table. Mark where it lands with tape. 2. **Measure:** How far from the table did the marble land? Record. 3. **Change the height:** Add 2 more books. Predict how far the marble will land NOW. 4. **Test:** Roll and measure. Was your prediction close? 5. **Repeat:** Try with 6 books. See the pattern? 6. **Graph it:** Create a bar graph showing ramp height vs. landing distance.

Math Connection: Calculate the difference between predicted and actual distances. Find the average landing distance for 3 trials at each height.

TEKS Connection: Math 4.8(A) Identify relative sizes of measurement units; Math 5.8(A-C) Represent and interpret data; Sci 5.1(E) Collect and record data.

STEM Challenge Card 4: Chien-Shiung Wu — Chain Reaction Challenge

Biography Side

Chien-Shiung Wu (1912-1997) was a Chinese American physicist who made groundbreaking discoveries about how atoms behave. She disproved a law of physics that scientists had believed for 30 years! Despite her incredible contributions, two male colleagues received the Nobel Prize for work she helped prove. She is often called the “First Lady of Physics.”

Challenge Side

The Chain Reaction Challenge

Your Mission: Set up a chain reaction using dominoes and simple objects where one action triggers the next — at least 5 steps!

Materials: - Dominoes or small wooden blocks (at least 20) - Marbles - Cardboard tubes (paper towel/toilet paper rolls) - Small cups - Ramps (books, cardboard) - String

Steps: 1. **Learn:** A chain reaction is when one event causes the next, like dominoes falling! 2. **Plan (10 min):** Draw your chain reaction. Label each step. You need at least 5

steps. 3. **Build (20 min):** Set up your chain reaction. Test sections as you go. 4. **Test (5 min):** Run the full chain reaction! Did every step trigger the next? 5. **Record:** Draw what happened. Identify which step failed (if any) and why. 6. **Fix & Retest (10 min):** Make improvements.

Science Connection: Energy transfers from one object to the next — that's how chain reactions work!

TEKS Connection: Sci 4.6(A) Differentiate between forms of energy; Sci 3.6(A) Explore different forms of energy.

STEM Challenge Card 5: Mary Jackson — Wind Tunnel Challenge

Biography Side

Mary Jackson (1921-2005) was NASA's first Black female engineer. She worked at the Langley Research Center in Virginia, where she analyzed data from wind tunnel experiments to help design better, faster airplanes. Like Katherine Johnson, she was one of the real-life heroes of *Hidden Figures*. She later worked to help other women and minorities at NASA get promoted into engineering positions.

Challenge Side

The Wind Tunnel Challenge

Your Mission: Design a car shape from clay or cardboard that travels the farthest when pushed by wind (a fan or blow dryer).

Materials: - Modeling clay or cardboard + tape - Small wheels (optional — or use round candies/beads) - A fan or blow dryer on cool setting - Meter stick or measuring tape - Flat, smooth surface (table or floor)

Steps: 1. **Research (5 min):** What shapes move through air easily? (Think: fish, airplanes, race cars) 2. **Design (5 min):** Sketch TWO different car shapes. Predict which will go farther. 3. **Build (15 min):** Build both designs. Keep them the same weight! 4.

Test (10 min): Place each car at the same starting point. Turn on fan for 5 seconds. Measure distance traveled. Do 3 trials each. 5. **Compare:** Calculate average distance for each design. Which shape was more aerodynamic?

TEKS Connection: Sci 5.1(B) Design simple experimental investigations; Math 5.3(K) Add and divide decimals; Sci 4.5(A) Measure and compare physical properties.

STEM Challenge Card 6: Wangari Maathai — Seed Germination Challenge

Biography Side

Wangari Maathai (1940-2011) was a Kenyan environmental scientist who founded the Green Belt Movement, planting over 51 million trees across Africa. She was the first African woman to win the Nobel Peace Prize (2004). She believed that caring for the environment and fighting for human rights go hand in hand. She earned a PhD in biology — the first woman in East and Central Africa to do so.

Challenge Side

The Seed Germination Challenge

Your Mission: Test which conditions help seeds sprout the fastest — and grow a tiny garden!

Materials: - Bean or radish seeds (fast growers!) - Small cups or plastic bags - Paper towels or cotton balls - Water - Plastic wrap - Ruler - Markers

Steps: 1. **Set up 4 groups** (3 seeds each): - Group A: Wet paper towel, warm spot, light - Group B: Wet paper towel, warm spot, NO light (cover with box) - Group C: Wet paper towel, COOL spot (near window in winter), light - Group D: DRY paper towel, warm spot, light 2. **Predict:** Which group will sprout first? Why? 3. **Observe:** Check every day for 7 days. Measure growth. Draw what you see. 4. **Record data:** Create a table and line graph of growth over time. 5. **Conclude:** What conditions helped seeds grow best?

TEKS Connection: Sci 4.9(A) Investigate the basic needs of organisms; Sci 3.9(A) Observe and describe life cycles; Math 4.9(A) Represent data on a frequency table or line plot.

STEM Challenge Card 7: Grace Hopper — Debugging Logic Challenge

Biography Side

Rear Admiral Grace Hopper (1906-1992) was a computer science pioneer who helped create one of the first programming languages (COBOL). She famously found an actual moth stuck in a computer, and the term “debugging” (fixing computer errors) became popular! Grace served in the U.S. Navy and was one of the oldest active-duty officers when she retired at age 79.

Challenge Side

The Debugging Logic Challenge

Your Mission: Find and fix the “bugs” in a series of logic puzzles and coded messages!

Activity 1 — Code Breaker: Each letter is replaced by a number (A=1, B=2, etc.). Decode the secret message: - 19-3-9-5-14-3-5 / 9-19 / 6-21-14! (Answer: SCIENCE IS FUN!)

Activity 2 — Pattern Bug: Find the error in each number pattern: - 2, 4, 6, 8, 11, 12, 14 (Bug: 11 should be 10) - 5, 10, 15, 25, 25, 30 (Bug: first 25 should be 20) - 100, 90, 80, 70, 55, 50 (Bug: 55 should be 60)

Activity 3 — Algorithm Fix: A recipe for making a sandwich has the steps in the wrong order. Put them in the correct sequence: 1. Put the two slices of bread together 2. Get two slices of bread 3. Add lettuce and tomato 4. Spread peanut butter on one slice 5. Eat the sandwich (Correct: 2, 4, 3, 1, 5)

Activity 4 — Create Your Own Code: Write a 3-step “program” (set of instructions) for a simple task. Then hide ONE bug in it. Trade with a partner to debug!

TEKS Connection: Math 3.5(A-E) Represent and solve problems with multiplication/division; Math 4.5(A) Represent multi-step problems.

STEM Challenge Card 8: Marie Curie — Crystal Growing Challenge

Biography Side

Marie Curie (1867-1934) was a Polish-French physicist and chemist who discovered two elements: polonium and radium. She was the first woman to win a Nobel Prize, and the ONLY person ever to win Nobel Prizes in TWO different sciences (physics and chemistry). Despite facing discrimination as a woman in science, Marie's discoveries changed medicine and physics forever.

Challenge Side

The Crystal Growing Challenge

Your Mission: Grow crystals using a supersaturated solution and observe how they form!

Materials: - Hot water (teacher pours!) - Sugar OR salt OR borax (teacher choice) - Glass jars - Pipe cleaners (bent into shapes) - String and pencils - Magnifying glass - Recording sheet

Steps: 1. **Set up (teacher helps with hot water):** Dissolve as much sugar/salt/borax as possible in hot water until no more dissolves. 2. **Prepare:** Bend pipe cleaner into a shape. Tie to pencil with string. Lower into solution. 3. **Wait:** Place jar where it won't be disturbed. Check every day! 4. **Observe (Days 1-5):** Use magnifying glass. Draw what you see each day. Measure crystal growth. 5. **Compare:** If different groups used different substances, compare crystal shapes and sizes. 6. **Reflect:** Why do crystals form as the water evaporates?

TEKS Connection: Sci 3.5(A) Measure, test, and record physical properties of matter; Sci 4.5(A) Measure and compare physical properties; Sci 5.5(A) Classify matter based on physical properties.

STEM Challenge Card 9: Sylvia Earle — Ocean Pressure Challenge

Biography Side

Sylvia Earle (born 1935) is a marine biologist and ocean explorer known as “Her Deepness.” She set a record for the deepest untethered walk on the ocean floor at 1,250 feet! Sylvia has spent over 7,000 hours underwater and led more than 100 expeditions. She has dedicated her life to protecting the world’s oceans and the creatures living in them.

Challenge Side

The Water Pressure Challenge

Your Mission: Investigate how water pressure changes with depth!

Materials: - Empty 2-liter plastic bottle - Nail or pushpin (teacher use only) - Tape - Water - Ruler - Tray or sink to catch water - Recording sheet

Steps: 1. **Set up (teacher pokes 3 holes):** One near the top, one in the middle, one near the bottom of the bottle. Cover holes with tape. 2. **Predict:** When you fill the bottle and remove the tape, which hole will shoot water the farthest? Why? 3. **Test:** Fill bottle with water. Remove tape from all 3 holes at once (do this over a tray!). 4. **Observe & Measure:** Measure how far water shoots from each hole. 5. **Explain:** The bottom hole shoots farthest because there’s MORE water above it pushing down = more pressure! 6. **Connect:** How does this relate to what deep-sea divers like Sylvia Earle experience?

TEKS Connection: Sci 3.6(A) Explore different forms of energy; Sci 5.6(B) Demonstrate that the flow of electricity requires a complete circuit (related force concepts).

STEM Challenge Card 10: Temple Grandin — Sensory Maze Challenge

Biography Side

Temple Grandin (born 1947) is an animal scientist, professor, and autism advocate. Diagnosed with autism as a child, she thinks in pictures — which gives her a unique ability to understand how animals see and experience the world. She revolutionized livestock handling by designing systems that reduce animal stress. Temple is a professor at Colorado State University and has shown the world that thinking differently is a superpower.

Challenge Side

The Sensory Maze Challenge

Your Mission: Design a maze that a blindfolded partner can navigate using ONLY touch and sound clues!

Materials: - Cardboard boxes or large pieces of cardboard - Textured materials (sandpaper, cotton, foil, bubble wrap, felt) - Bells, shakers, or crinkly materials - Tape - Blindfold (bandana) - Timer

Steps: 1. **Learn:** Temple Grandin thinks about how animals experience spaces through their senses. You'll design a space experience too! 2. **Plan (10 min):** Design a tabletop maze path (at least 3 turns). Plan texture clues: "When you feel sandpaper, turn right. When you feel cotton, turn left." 3. **Build (20 min):** Construct your maze with tactile clue markers at each turn. 4. **Write instructions:** Create a "sensory key" card for your partner. 5. **Test (10 min):** Blindfold a partner. Give them the key. Can they navigate the maze? Time them! 6. **Reflect:** How did it feel to rely on senses other than sight?

TEKS Connection: Sci 3.1(A) Ask questions based on observations; Sci 4.1(A) Ask questions and plan investigations; Sci 5.1(G) Develop explanations using evidence.

Student Recording Sheet (Universal Template)

Name: _____ **Date:** _____

Challenge Card #: _____ **Woman in STEM:** _____

My Prediction

I think _____ will happen because _____

My Design (sketch here)

[Drawing space]

My Data

Trial	Measurement/Observation
1	
2	
3	

What I Changed & Why

Results After Improvement

Reflection

1. What worked well? _____
 2. What was challenging? _____
 3. What would you do differently next time? _____
 4. How does this connect to [woman's name]'s real work? _____
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Google Slides Version Outline

Slide 1: Title slide — “Women’s History Month STEM Challenge Cards”

- Purple/teal gradient background with gear and beaker icons

Slides 2-11: One slide per woman (biography display)

- Left: Biography text, Right: Placeholder for image/illustration
- Fun facts callout box
- “Did You Know?” bubble

Slides 12-21: Challenge instruction slides (one per challenge)

- Step-by-step with numbered icons
- Materials list with checkboxes
- Timer display graphic

Slides 22-31: Digital recording sheets (one per challenge)

- Editable text boxes for predictions, data, reflections
- Table for data entry

Slide 32: STEM Challenge Certificate

- “I completed ____ STEM challenges!” with stars and name field

Slide 33: Vocabulary Review

- Key STEM vocabulary with definitions